

# MMO - Black Box

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# Function 7

## The Optimization Problem



- Binary Search Space: 30 Variables
  - $2^{30}$  -> 1 billion possible solutions
- Bi-Objective minimization (f1, f2)
- Black-Box Problem
  - No gradient information
  - No known true pareto front
- Reference Point [10.000, 10.000]

# Function 7

## Search Space Characterization



- Concave front shape
  - Steep decline
- Strong trade-off between  $f_1$  and  $f_2$
- Large range of  $f_2$ 
  - Larger optimization potential

# Experimental Setup

Fair Evaluation/Comparison



| Parameter                       | Value                                 |
|---------------------------------|---------------------------------------|
| Algorithms                      | NSGA-II, SMS-EMOA, MOEA/D             |
| Population Size                 | 100                                   |
| Generations                     | 100                                   |
| Runs per Algorithm              | 5                                     |
| Operators                       | HalfUniformCrossover, BitflipMutation |
| Seeds                           | 42, 142, 242, 342, 442                |
| → Total of ~150.000 evaluations |                                       |

# Quality Metrics

How to measure Quality?



## Hypervolume

- Measures volume between reference point and pareto-front
  - Convergence + diversity
- Pareto-compliant indicator
- Set A dominates B

## Inverted Generational Distance

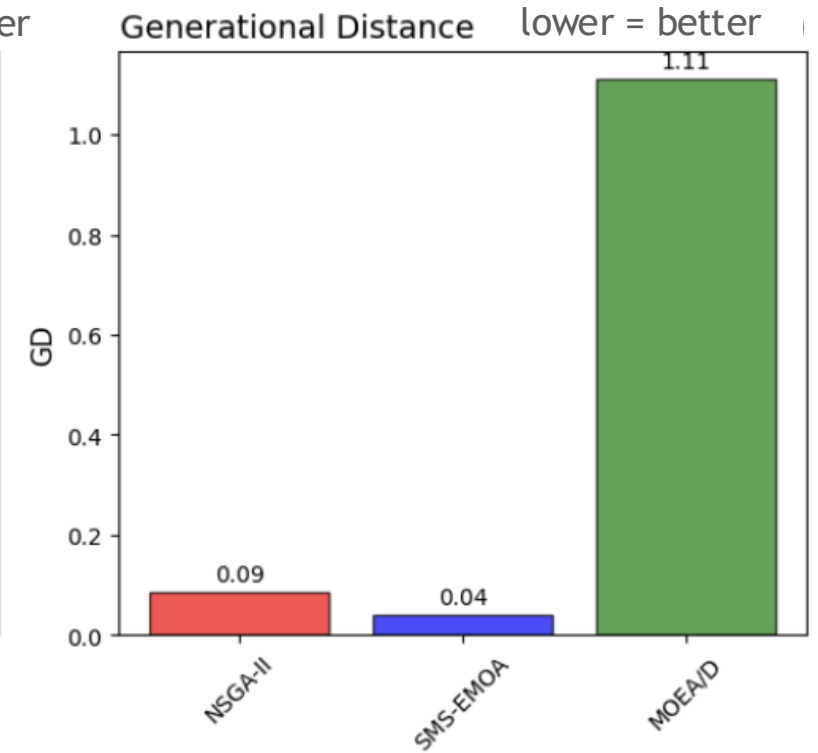
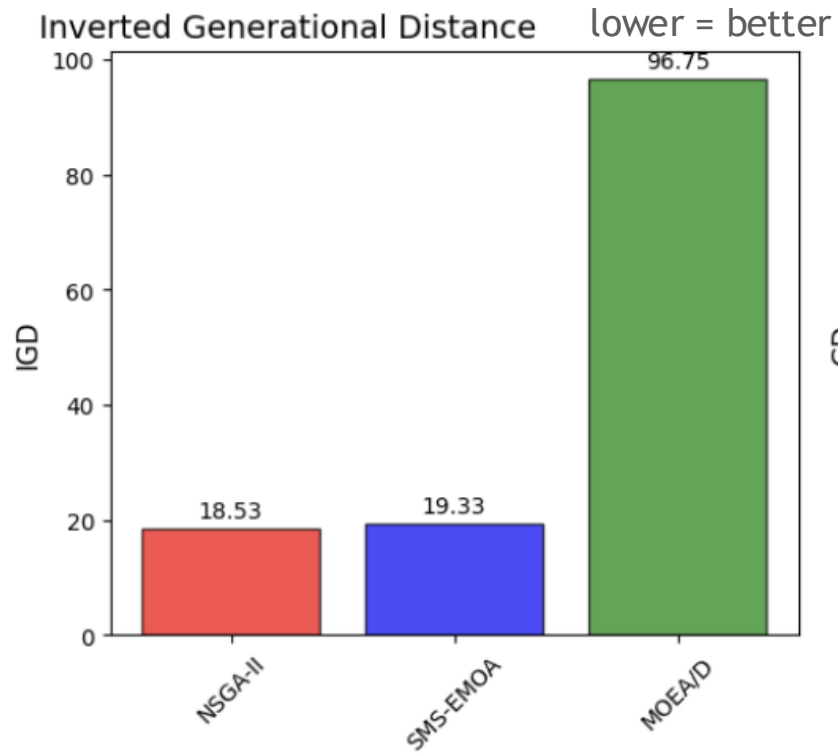
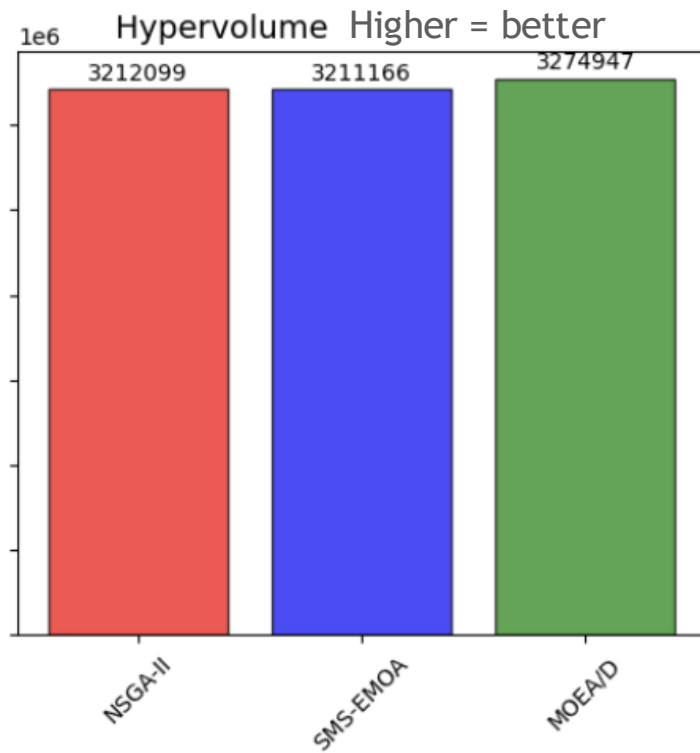
- Average distance from reference set points to nearest approximated solution
- Convergence + diversity
- Requires known reference front
- Sensitive to missing regions in the front

## Generational Distance

- Average distance from approximated solutions to nearest reference point
- Primarily convergence
- Not penalizing poor diversity

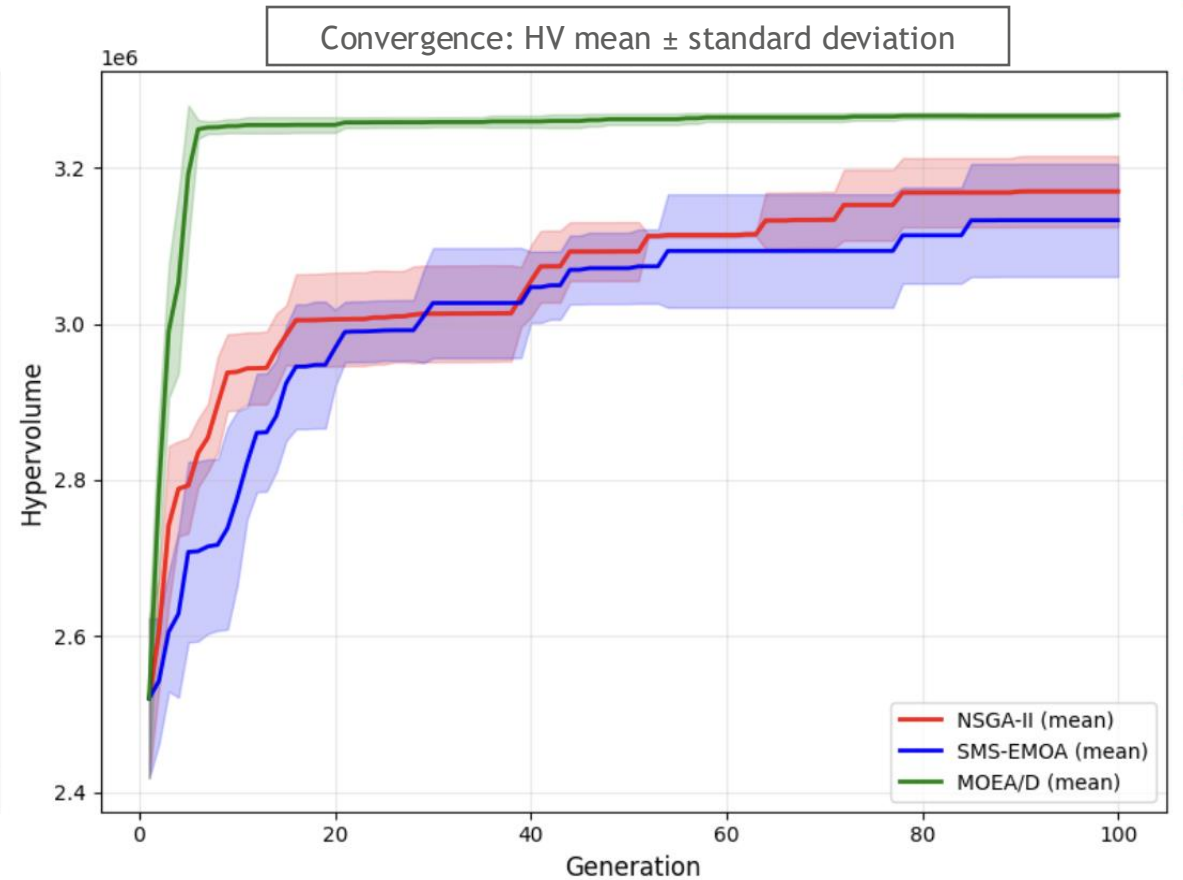
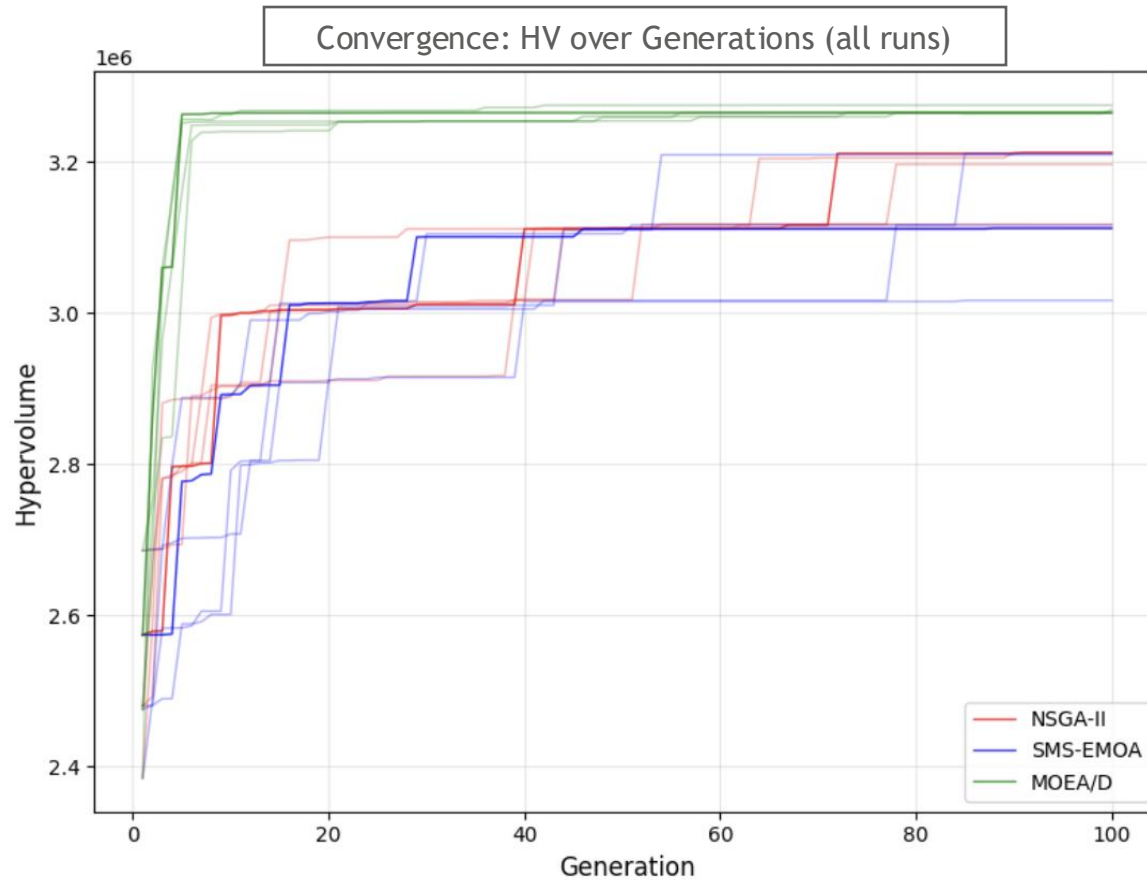
# Results

## Quality Metrics



# Results

## Convergence Behavior

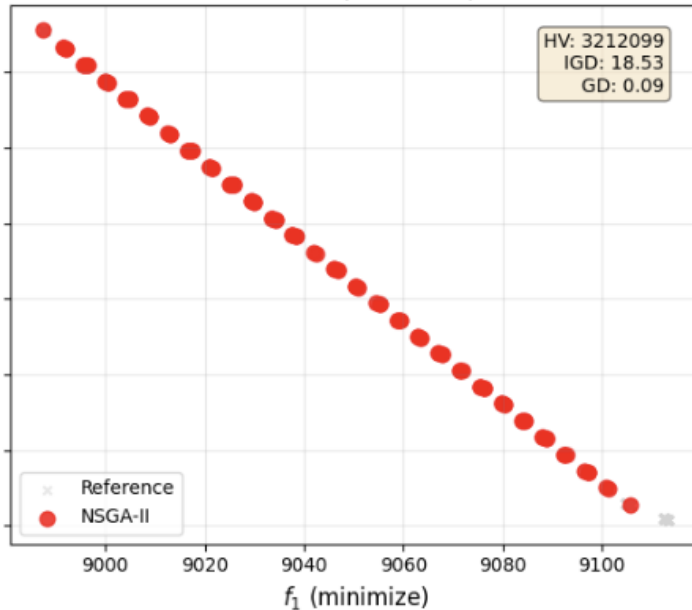


# Results

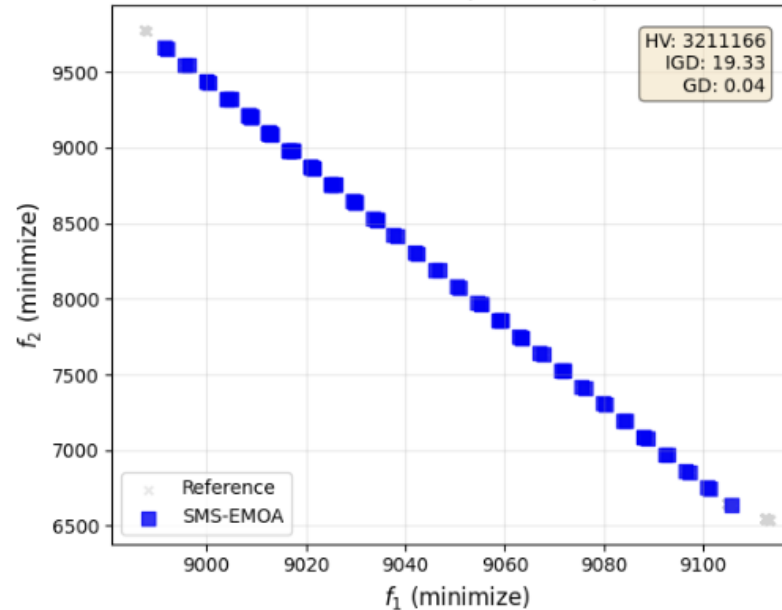
## Pareto-Front



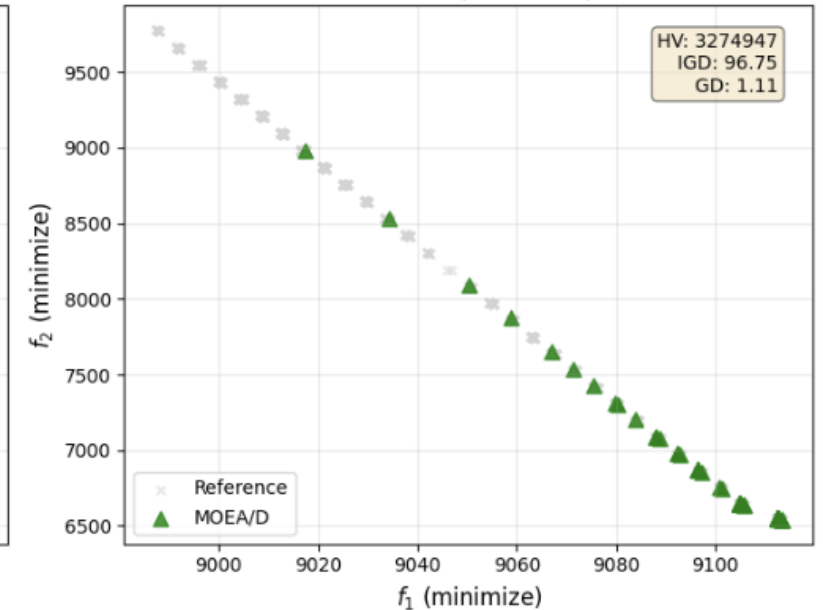
NSGA-II (Best Run)



SMS-EMOA (Best Run)



MOEA/D (Best Run)



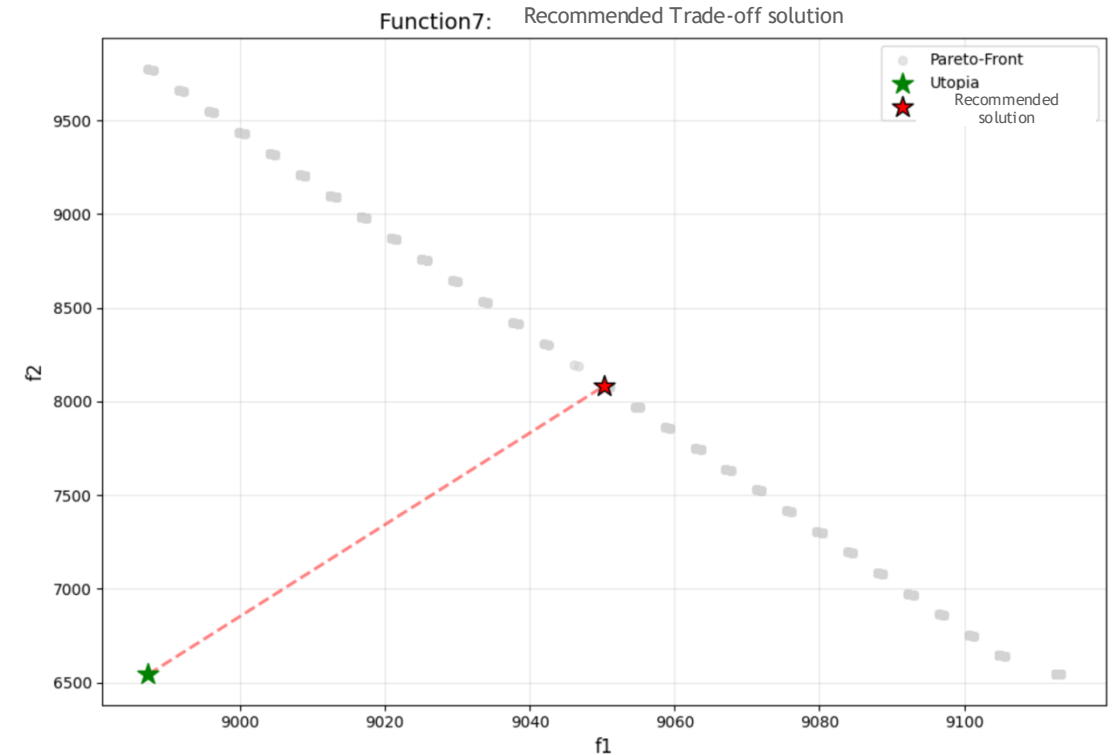
- 1.342 non-dominated solutions (combined)
- $f_1$  range [8.987 - 9.113]
- $f_2$  range [6.542 - 9.777]



# Results

## Recommended Solution

- Minimal normalized distance to Utopia-point
- Utopia:  $f1_{\min}$ ,  $f2_{\min}$ 
  - Theoretical unreachable
- Solution:  $f1 = 9050.29$ ,  $f2 = 8083.76$





**Thank you for your attention!**